

Seat. No. _____

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BE SEMESTER-VIII (NEW) Examination July – 2025
Subject Name: Next Generation Networks

Subject Code: CE801-N

Date: 14/07/2025

Time: 10:30am to 01:30pm

Total Marks: 70

Instructions:

1. Answer each section in separate answer sheet.
2. Use of scientific calculator is permitted.
3. All questions are Compulsory.
4. Indicate clearly, the option you attempt along with its respective question number.
5. Use the last page of main supplementary of rough work.

Section-I

- Q-1 (A)** What are the essential functional differences between 1st generation, 2nd generation and 3rd generation. [5]
- (B)** Describe architecture of mobile computing with figure in detail.. [5]
- (C)** Explain Mobility management with respect to GSM. [5]
- OR**
- (C)** What is mobile computing? Give example of application of Mobile computing. [5]
- Q-2 (A)** What is multiplexing? Explain types of multiplexing in details. [5]
- (B)** Draw the GPRS system architecture, and explain function of each block.. [5]
- OR**
- Q-2 (A)** Discuss the security features in UMTS. [5]
- (B)** What is WAP? Describe WAP architecture and its importance. [5]
- Q-3 (A)** Explain DSDV and DSR protocols with examples. [5]
- (B)** Explain GSM architecture with figure. [5]
- OR**
- Q-3 (A)** Explain the concept and working of ODMRP. [5]
- (B)** Compare AODV and ZRP routing protocols.. [5]

Section-II

- Q-4** (A) What is Mobile TCP? Explain it in detail. [5]
(B) Explain Standards-Developing Organizations. [5]
(C) What is GPRS in detail? Explain its characteristics, application and security. [5]

OR

- (C) Write down SDN and NFV Open Standards Activities. [5]
- Q-5** (A) Explain Mobility management with respect to GPRS. [5]
(B) How does mobile IP work? What are the challenges with mobile IP with respect to High-speed mobility? How does cellular IP solve some of these challenges? [5]

OR

- Q-5** (A) Explain importance and significance of Mobile IP. [5]
(B) Explain MANET in details. [5]

- Q-6** (A) Explain VANET in details. [5]
(B) Describe the Open Flow logical switch architecture. [5]

OR

- Q-6** (A) What is role of data center Networking.? [5]
(B) Explain three most significant efforts in Open Development Initiatives. [5]

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